

PROJECT REPORT – EMOTIONS DETECTION FROM TEXT

Internship Project Title	RIO-45:Automate detection of different emotions from textual comments and feedback
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Name of the Company	TCS iON
Name of the Industry Mentor	Debashis Roy
Name of the Institute	Rajalakshmi Engineering College

Start Date	End date	Total no. of hours	Project Environment	Tools used
27-07-2021	06.08.2021	20 Hours	Google colab	Datasets using python

PROJECT SYNOPSIS

ABSTRACT

Emotion detection (ED) is a branch of sentiment analysis that deals with the extraction and analysis of emotions. The evolution of Web 2.0 has put text mining and analysis at the frontiers of organizational success. It helps service providers provide tailor-made services to their customers. Numerous studies are being carried out in the area of text mining and analysis due to the ease in sourcing for data and the vast benefits its deliverable offers. In today's world communication is done on vast area along with growth of Internet. Communication and human interaction is carried out using social sites, applications etc. in textual form. Emotion plays a vital role in human day-to-day life conveying their feelings and situation in their life. Emotions can be expressed using speech, gestures facial expression, or text. Emotions includes happy or joy, sad, disgust, anger, fear and surprise as basic emotion. Emotion Detection in text documents is essentially a content – based classification problem involving concepts from the domains of Deep learning. In this paper emotion detection based on textual data and the model used in it are discussed here.

ACKNOWLEDGEMENT

The internship opportunity I have with TCS iON was a great chance for learning and developing my programming skills. Therefore I consider myself as lucky individual as I was provided to be part of it.

And I am so grateful for having chance to communicate with wonderful people and professionals who led me through this internship period.

I would like to use this opportunity to express my deepest sense of gratitude to my industry mentor Mr. Debashis Roy who heartily welcomed me for the internship and guided and encouraged me throughout the internship.

Also, I express my sincere thanks to my academic mentor Ms. Jinu Sophia for constant support and who were kind enough to give opportunity to work under their immense expertise and for giving me valuable suggestions, motivation, encouragement and support towards this project.

Last but not the least. I am thankful and indebted to all those who helped me directly and indirectly in this project for their careful and precious guidance which were extremely valuable for my project in both theoretically and practically which led to completion of project report.

The study has indeed helped me to explore more in knowledge avenues related to my topic. I am sure it will help me in my future.

OBJECTIVE

To develop a deep learning algorithm to detect the different types of emotion contained in a collection of English sentences or a large paragraph

INTRODUCTION

Emotion is one type of affect, other type of being mood, temperament and sensation. Emotions have been widely studied in psychology and in behavior sciences, as they are an important element of human nature. Nowadays they have also attracted the attention of researchers of computer science, especially in the field of human computer interactions. Advancement in textual analysis have allowed the area of emotion detection to become a recent interest in computational linguistic. Emotion detection is the newer area of textual analysis and therefore, has weaker standard methods. Emotion can be expressed as happiness, sadness and so forth. While board topic of emotion has been studied in psychology for decades, very little effort has been spent on attempting to detect emotion from text. In this work, we assume that emotion reaction of an input sentence is essentially represented by its word appearance. Emotion Detection from text is a recent field of research that is closely related to Sentiment Analysis. Sentiment Analysis aims to detect positive, neutral, or negative feelings from text, whereas Emotion Analysis aims to detect and recognize

types of feelings through the expression of texts, such as anger, disgust, fear, happiness, sadness, and surprise. Emotion detection may have useful applications, such as:

- Gauging how happy our citizens are. Different indexes have different definitions; most evolve around economic, environmental, health, and social factors. Since the mid-2000s, Government and organizations around the world are paying increasing attention to the happiness index.
- Pervasive computing, to serve the individual better. This may include suggesting help when anxiety is detected through speech, or to check the tone of an email before sending it out.
- Understanding the consumer. Improving perception of a customer with the ultimate goal to increase brand reputation and sales.

Deep learning is a set of learning methods attempting to model data with complex architectures combining different non-linear transformations. The elementary bricks of deep learning are the neural networks that are combined to form the deep neural networks. These techniques have enabled significant progress in the fields of sound and image processing, including facial recognition, speech recognition, computer vision, automated language processing, text classification (for example spam recognition). Potential applications are very numerous. A spectacularly example is the AlphaGo program, which learned to play the go game by the deep learning method, and beat the world champion in 2016. There exist several types of architectures for neural networks:

- The multilayer perceptrons, that are the oldest and simplest ones
- The Convolutional Neural Networks (CNN), particularly adapted for image processing
- The recurrent neural networks, used for sequential data such as text or times series.

Neural Networks are set of algorithms which closely resemble the human brain and are designed to recognize patterns. They interpret sensory data through a machine perception, labeling or clustering raw input. They can recognize numerical patterns, contained in vectors, into which all real-world data (images, sound, text or time series), must be translated. Artificial neural networks are composed of a large number of highly interconnected processing elements (neuron) working together to solve a problem. An ANN usually involves a large number of processors operating in parallel and arranged in tiers. The first tier receives the raw input information — analogous to optic nerves in human visual processing. Each successive tier receives the output from the tier preceding it, rather than from the raw input — in the same way neurons further from the optic nerve receive signals from those closer to it. The last tier produces the output of the system..

Recurrent Neural Network is a generalization of feedforward neural network that has an internal memory. RNN is recurrent in nature as it performs the same function for every input of data while the output of the current input depends on the past one computation. After producing the output, it is copied and sent back into the recurrent network. For making a decision, it considers the current input and the output that it has learned from the previous input. Unlike feedforward neural networks, RNNs can use their internal state (memory) to process sequences of inputs. This makes them applicable to tasks such as unsegmented, connected handwriting recognition or speech recognition. In other neural networks, all the inputs are independent of each other. But in RNN, all the inputs are related to each other.

Long Short-Term Memory (LSTM) networks are a modified version of recurrent neural networks, which makes it easier to remember past data in memory. The vanishing gradient problem of RNN is resolved here. LSTM is well-suited to classify process and predict time series given time lags of unknown duration. It trains the model by using back-propagation.

INTERNSHIP ACTIVITIES

First, we are supposed to attend pre – test first and after completing with target we have learn and understand the reference materials. After that we need go through the industry project and simultaneously we should the daily day report in provided flat form and in the same we should work with our project. After completing the project we should upload the project report and attend the project test and at last with the help of academic mentor we should submit the viva report.

Project is about to identifying the different types of emotions in dataset or a large paragraph using deep learning with certain conditions.

APPROACH

Bi-LSTM - Bidirectional – Long Short Term Memory

Bidirectional long-short term memory (bi-lstm) is the process of making any neural network o have the sequence information in both directions backwards (future to past) or forward (past to future). In bidirectional, our input flows in two directions, making a bi-lstm different from the regular LSTM. With the regular LSTM, we can make input flow in one direction, either backwards or forward. However, in bi-directional, we can make the input flow in both directions to preserve the future and the past information.

ASSUMPTIONS

We assume that any random individual would generally agree on the emotion labels associated with any given sentence or word. This is the most problematic assumption. It's believed that the emotion detectors learn the individual-specific criteria to tag sentiments in texts, but don't generalize well from one human tagger to next. We can assume that language doesn't change and neither the meaning of words. This is true, if we study corpus which are collected over a relatively short period of time or a decade. However, some heuristics and tricks allow us to preserve them and thus conduct emotion detection analysis.

EXCEPTIONS

- Using Ambiguity in Keywords Definitions

Using emotions keywords in straightforward way to detect associated emotions, the meanings of keywords could be multiple and vague as most words could change their meanings according to different usages and contexts.

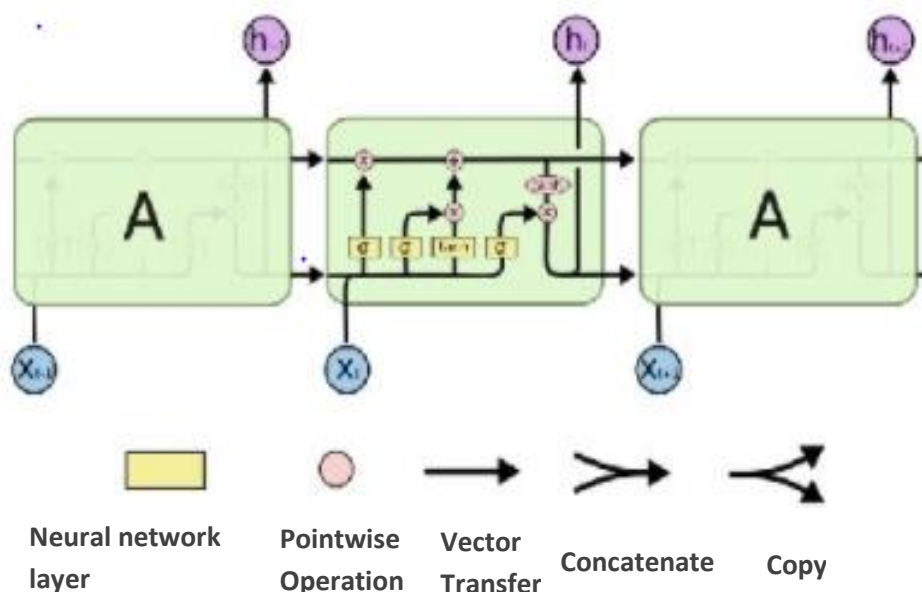
- Lack of Linguistic Information

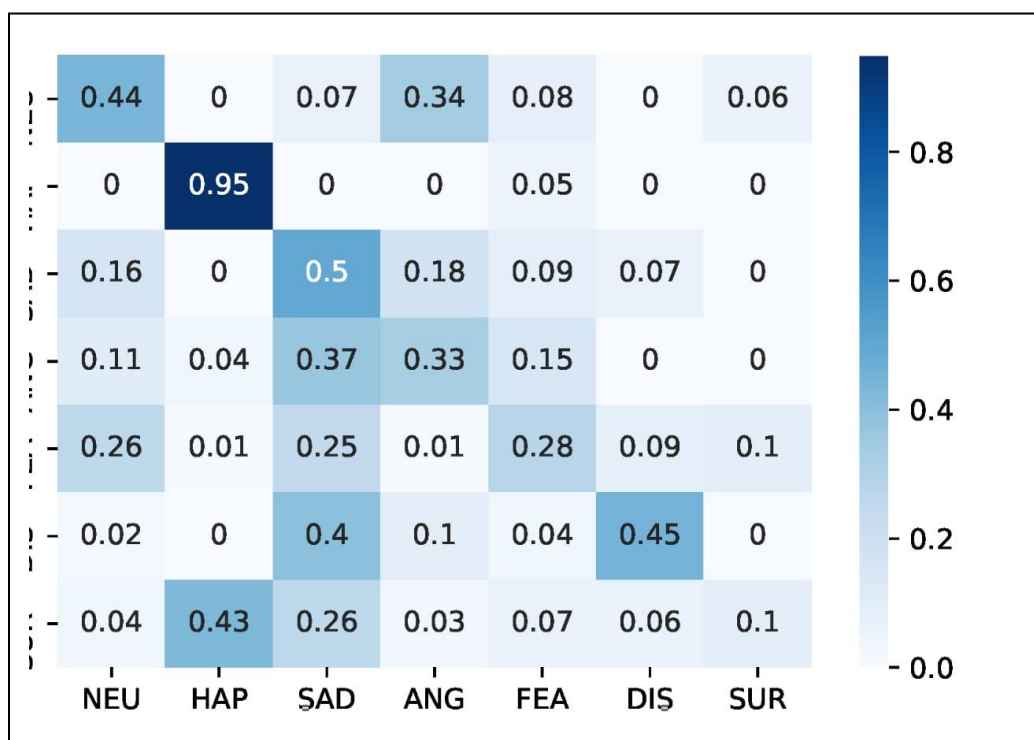
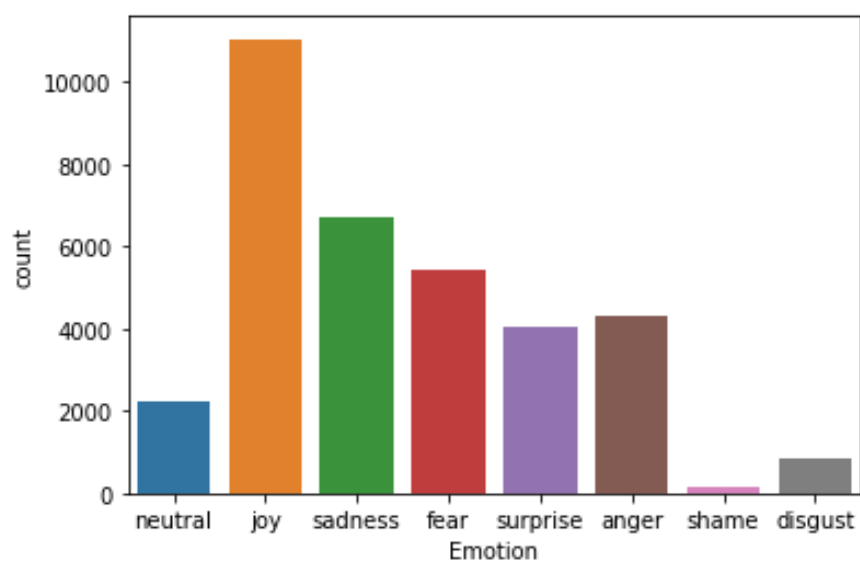
Syntax structures and semantics also have influences on expressed emotions.

- Difficulties in Determining Emotion Indicators

Learning-based methods can automatically determine the probabilities between features and emotions but the methods still need keywords, but in the form of features.

DIAGRAMS, CHARTS & TABLES - LSTM DIAGRAM





DATA SET



	Text	Emotion
0	i didnt feel humiliated	sadness
1	i can go from feeling so hopeless to so damned...	sadness
2	im grabbing a minute to post i feel greedy wrong	anger
3	i am ever feeling nostalgic about the fireplac...	love
4	i am feeling grouchy	anger

DATA SET SADNESS

	Text	Emotion	Sentiment
0	i didnt feel humiliated	sadness	Neutral
1	i can go from feeling so hopeless to so damned...	sadness	Neutral
5	ive been feeling a little burdened lately wasn...	sadness	Positive
10	i feel like i have to make the suffering i m s...	sadness	Negative
13	i feel low energy i m just thirsty	sadness	Neutral
...	...	...	...
20731	The tears came , and rapidly dissolved her ang...	sadness	Negative
20732	Through a swell of tears Ruth stared incredulo...	sadness	Neutral
20733	She glanced down at the linoleum in despair	sadness	Negative
20734	Perhaps it was despair that made me believe th...	sadness	Neutral
20735	That he forgot his guest in his despair	sadness	Neutral

6265 rows x 3 columns



DATA SET ANGER

	Text	Emotion	Sentiment
2	im grabbing a minute to post i feel greedy wrong	anger	Negative
4	i am feeling grouchy	anger	Neutral
12	i think it s the easiest time of year to feel ...	anger	Neutral
20	i feel irritated and rejected without anyone d...	anger	Neutral
24	i already feel like i fucked up though because...	anger	Negative
...	...	...	...
21175	Irina hung up in exasperat	anger	Neutral
21176	She was acting like some idiotic schoolgirl , ...	anger	Positive
21177	Jean-Paul gave a sigh of exasperat	anger	Neutral
21178	Parkin sighed with exasperat	anger	Neutral
21179	With a small sigh of exasperation she thrust h...	anger	Negative

2993 rows x 3 columns

DATA SET FEAR

	Text	Emotion	Sentiment
7	i feel as confused about life as a teenager or...	fear	Negative
19	i now feel compromised and skeptical of the va...	fear	Negative
21	i am feeling completely overwhelmed i have two...	fear	Positive
31	i remember feeling acutely distressed for a fe...	fear	Positive
53	i was stymied a little bit as i wrote feeling ...	fear	Negative
...	...	...	...
21454	Melissa stared at her friend in dism	fear	Neutral
21455	Successive state elections have seen the gover...	fear	Neutral
21456	Vincent was irritated but not dismay	fear	Neutral
21457	Kendall-Hume turned back to face the dismayed ...	fear	Neutral
21458	I am dismayed , but not surpris	fear	Neutral

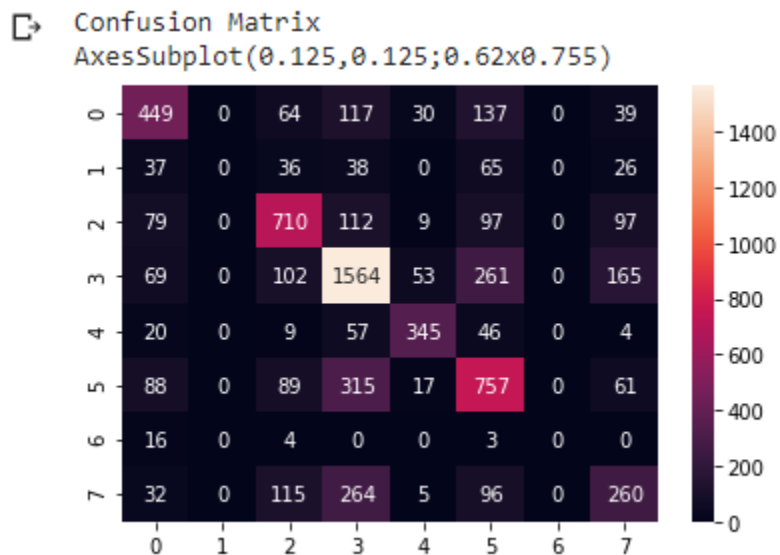
2652 rows x 3 columns

DATA SET – SURPRISE

	Text	Emotion	Sentiment
6	ive been taking or milligrams or times recomme...	surprise	Positive
32	i have seen heard and read over the past coupl...	surprise	Positive
57	i keep feeling pleasantly surprised at his sup...	surprise	Positive
64	i am now nearly finished the week detox and i ...	surprise	Positive
129	i too feel as if i am a stranger in a strange ...	surprise	Negative
...	...	...	...
20891	If the birds are disturbed during the process ...	surprise	Positive
20892	It was Blackberry who bullied the stupefied Pi...	surprise	Negative
20893	He woke stupefied , flicking his tongue over h...	surprise	Neutral
20894	Laura stared up at Ross in stupefied ama	surprise	Neutral
20895	The pain had faded into the background and she...	surprise	Neutral

879 rows x 3 columns

CONFUSION MATRIX



ALGORITHMS:

LINK 2 – EMOTION CODE

START

Step 1: Setting up the Development Environment

- Open Google Colab and connect the server

Step 2: Choosing the Dataset

- Import numpy and pandas
- Upload Dataset in google Colab and run the csv file

Step 3: Emotion count

- Count the value of emotions

Step 4: Sentimental Analysis

- Import TextBlob
- Apply cases of Positive, Negative, Neutral in if-else loop
- Enter the text and identify the statement it is Positive, Negative or Neutral
- Test the function

Step 5: emotions vs sentiments

- Compare emotions and sentiments
- List of emotions
- And then run the each emotion using code.

Step 6

- Load the packages of vectorizer, metrics and split the data.
- Build the features of text
Count the vectorizer by name and atlast split the dataset by train and test.

Step 7: Build our model

- Load new view model (nv model)
- Add accuracy and predictions.
- Check the array

Step 8: Vectorized text and apply model

- Enter the sample text and vectorize it.
- Make prediction and check the probability.
- Predict the emotions of model and nv model and run the sentence.

Step 9: Model Evaluation

- Print the classification.
- Test the confusion matrix
-

Step 10: Save Model

- Import joblib.
- Upload the pkl file and run it.
- Load logistic Regression model
- Add accuracy, score and predict the emotions
END

LINK 1 – EMOTION

START

Step 1: install the python package text2emotion.

Step 2: import text2 emotion.

Step 3: enter the text or large paragraph.

Step 4: run the text and identify the different types of emotion.

END

LINK 3 – LSTM MODEL

START

Step 1: open google colab and connect the server.

Step 2: Choosing the dataset

- Import packages
- Upload dataset in google colab

- And also embedding files in dataset
- Then plot a target class graph

Step 3: Then download nltk package file.

Step 4: calculate mean and standard for the pre trained weights.

Step 5: Find the max words and padding the sequences.

Step 6: Creating model and plot the graph

- Load the weights
- Calculate accuracy code
- And confusion matrix

Step 7: Test on own

- Map the target classes using label – encode
- And predict the emotions for happy, sad, anger, fear, joy.

STOP

CHALLENGES AND OPPORTUNITIES

CHALLENGES

The datasets are labeled by the people, and different persons can read and interpret emotions in different ways. Also, some visible visual cues like furrowed eyebrows can mean other emotions aside from anger, and other cues may be subtle hints of anger, although they are not crystal clear. When detecting emotions from people of different colors. There are models that detect more anger in black people. This means that training sets need to be more diverse, and experts are already doing what they can to fix this.

OPPORTUNITIES

The use of text-based ED had not been adequately explored for certain critical or life-saving applications. These areas include but not limited to crime detection and mitigation by analyzing messages of victims in order to identify threatening words, analysis of patient messages in order to determine depression levels of patients so that timely support could be offered, etc.

RISKS Vs REWARDS

It is problematic not just because it underplays the importance of cultural differences, but also because it assumes that there is a correlation between someone's facial expression and their emotional state. It is problematic to make inferences about someone's emotional state. But we already use a variety of techniques to do this in many aspects of society: personality tests are relatively common in work environments

REFLECTIONS ON THE INTERNSHIP

After doing this internship and project I got to know what is deep learning and how it is used, its application and different types of fields it is used and many approaches. And got know about emotion detection that is how different types of emotion are identified in large paragraph, sentence and a text

and also in social Medias like twitter. Learnt about different types of neural networks in deep learning and how to develop a model using dataset and models like LSTM and about Recurrent Neural Network and its classification.

Recommendations

- In Health Care.
- Automotive industry and emotion detection.
- In video game testing.
- Facial landmarks and in twitter to identify the emotions.
- I like to recommend others social Medias like Instagram, Facebook, etc.

Conclusion

Emotion Detection can be seen as an important field of research in human-computer interaction. A Sufficient amount of work has been done by researchers to detect emotion from facial and audio information whereas recognizing emotions from textual data is still a fresh and hot research area.

Enhancement scope

Emotion recognition has wide scope in many areas such as **human computer interaction**, biometric security etc. So it provides insight into artificial intelligence or machine intelligence that uses various supervised and unsupervised machine-learning algorithms to simulate the human brain.

Link to code and execution file

- Emotion
<https://colab.research.google.com/drive/1IMGooSHsr5u--kafW8ER0-TTgMP-JtNL?usp=sharing>
- Emotion code
<https://colab.research.google.com/drive/1ARxdF1H15mTKFZcXLYZ0Gg3f0UEcVKDI?usp=sharing>
- LSTM model
<https://colab.research.google.com/drive/16zZgja790OXWiz7Xw8or7BV0QXHtxVTb?usp=sharing>

RESEARCH PAPERS and RESPONSES

1. Has CNN taken over RNN in Speech Emotion Recognition? If yes, why?

CNN performs better than RNN at Image Processing and Speech Emotional Recognition(generally Speech Recognition)due to application of filters and Max Pooling which leads to elimination of lighter pixels in Image Processing(compared to darker); and noise and ... (compared to the voice phonemes)in Speech Recognition; accordingly , CNNs work by reducing an image/speech to its key

features and using the combined probabilities of the identified features appearing together to determine a classification. While RNNs deal with, only, sequential and there is no elimination and filtering which has made RNN fall back compared to CNN in aforementioned tasks.

2. What neurons to be fired to identify positive emotions in an neural network?

We are trying to build a cnn model to classify emotions from music. we know that only certain neurons will be triggered while identifying positive and negative emotions. But we cannot identify the neurons and a way to trigger them.

3. What are the algorithms/ tools / software packages do you recommend for Social Media Analytics?

Social media is one of the top distribution channels for user created content. By analyzing social media platform user shares, like tweets, photos, likes, re-tweets etc. one can see which types of posts receive the most engagement and use that rich information to form marketing, communication, channel and content strategy.

END OF THE REPORT